

MODULE- IV

POLYMERS AND GREEN FUELS

Polymers: Introduction, Molecular weight - Number average, weight average and numerical problems. Conducting polymers – synthesis and conducting mechanism of polyacetylene and commercial applications. Preparation, properties, and commercial applications of graphene oxide.

Green Fuels: Introduction, construction and working of solar photovoltaic cell, advantages, and disadvantages. Generation of energy (green hydrogen) by electrolysis of water and its advantages.

Self-learning: Regenerative fuel Cells. (8 hours)

POLYMERS

INTRODUCTION TO POLYMERS

The term "polymer" is derived from the ancient Greek word polus (meaning "many, much") and meros (meaning "parts"), and refers to a molecule whose structure is composed of multiple repeating units. The term was coined in 1833 by Jons Jacob Berzelius. Unmatched in the diversity of their properties, polymers such as cotton, wool, rubber, teflon and all plastics are used in nearly every industry. Natural and synthetic polymers can be produced with a wide range of stiffness, strength, heat resistance, density, light weight, corrosion resistance, high strength-to-weight ratio, electrical resistance, high wearing and abrasion resistance and at lower price. Because of these wonderful properties, polymers are being used extensively in clothing, home furnishing, food packaging, transportation, information technology, medical devices, coatings, electronics, engineering and structural materials to name a few.

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Natural polymers like silk, wool and cotton which have been used for thousands of years are replaced long ago by synthetic polymers like polyolefins, polyesters, nylon etc., in making of protective materials and in clothing. Many natural and synthetic polymers known as biomaterials, are extensively used in making artificial organs, bone

cements, dental cements, ligaments, pacemakers, contact lenses etc., as they are inert and not rejected by the body.

Polymer composites possess the properties required by modern technology such as aircraft applications. They have combination of unusual properties such as low density, high strength, stiffness, good abrasion, and impact and corrosion resistance. Any single metal, alloy, ceramic or polymeric material cannot offer the combination of the above properties. Typical uses of composites are in aerospace, automobiles, fishing rods, rockets, and bicycles. The **stealth** bomber was the first all-composite aircraft, but many passenger aircraft like **Airbus** and the **Boeing 787** use an increasing proportion of composites. Lot of research is going on in the field of conducting polymers which are already being used in many applications. They are used as electrodes for batteries, humidity sensors, gas sensors, conductive track on PCBs etc.

With continued research into the science and applications of polymers, they are playing an ever increasing role in society. In the coming years, polymers may replace many more conventional materials for various applications. In this chapter, we will have an overview of the world of polymers with a focus on future possibilities.

A polymer is a compound of high molecular weight made by the combination of large number of small repeating chemical units called as monomers.

Example: Polyethylene, Polystyrene etc.

As polymers are single, giant molecules, they are also called as macromolecules.

Monomers: These are the simple molecules, which combine with each other to form polymers. Monomers are also called as building blocks of polymers.

Example: Ethylene, vinyl chloride etc.

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Polymerization: The process of linking up of monomers to form polymers with or without the elimination of byproducts is called as polymerization.

Degree of polymerization (n): The total number of monomers present in a single chain of polymer is called as degree of polymerization.

When the value of n is very large, that is, in the range of hundreds or thousands, the polymers are called as high polymers. If value of n is less than 10, the polymers generally called as oligomers.

Functionality: The total number of bonding sites or functional groups present in a monomer is known as the functionality of the monomer.

For example; ethylene (bifunctional), styrene (bifunctional), phenol (trifunctional), etc.

CLASSIFICATION OF POLYMERS

Polymers are classified into different categories based on variety of parameters.

I. Based on the origin, polymers are classified as:

1. **Natural polymers:** Polymers found in nature, mostly in plants and animals are called as natural polymers.

Example: Natural rubber, silk, cotton etc.

2. **Semi Synthetic polymers:** These are mostly derived from naturally occurring polymers by carrying out chemical modifications. For example,



Cellulose diacetate is used in making threads, films, glasses etc.

3. **Synthetic polymers:** The polymers which are synthesized in the laboratory are called as synthetic polymers.

Example: Polythene, synthetic rubber, PVC, nylon etc.

Based on the composition, polymers are classified as:

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Homopolymers: Polymers which are made by the combination of only one kind of monomer are called as homopolymers.

Example: Polythene, PVC, polystyrene, Teflon etc.

Copolymers: Polymers which are made by the combination of more than one kind of monomers are called as copolymers or mixed polymers.

Example: Bakelite, nylon, buna-S rubber, polyurethanes etc.

III. Based on the method of preparation, polymers are classified as:

Addition polymers: The polymers formed by the addition of several monomers without elimination of byproducts are called as addition polymers.

Example: Polythene, PVC, Teflon etc.

Condensation polymers: The polymers formed by reaction between several monomers with elimination of simple molecules like, water, ammonia or methanol, etc., are called condensation polymers.

Example: Nylon, polyurethane, Bakelite

IV. Based on their behavior to heat treatment, polymers are classified as:

Thermoplastic polymers: The polymers which soften on heating, which can be converted into any shape and harden on cooling are called thermoplastic polymers. The process of heating, reshaping and cooling can be repeated number of times without affecting their properties.

Example: Polyethylene, polymethylmethacrylate, etc.

2. Thermosetting polymers: These are the polymers which undergo chemical changes on heating and become permanently hard, rigid, infusible on cooling.
Example: Phenol – formaldehyde, urea – formaldehyde, etc.

V. Based on their structure, polymers are classified as:

Linear polymers: These are the polymers in which monomeric units are linked together to form long straight chains. The polymeric chains stacked over one another to form a well packed structure. As a result of close packing, such polymers have high

density, high tensile strength and high melting points.

Example: High density polythene, nylons and polyesters.

All thermoplastics are essentially linear molecules.

Branched chain polymers: These polymers contain molecules having a linear backbone with branches arising randomly from it. Branched chain polymers are irregularly packed and thus, they have lower densities, tensile strength and melting points.

Example: LDPE, polystyrene, PMMA etc.

Cross-linked polymers: These polymers contain molecules having number of straight chains of varying lengths interlinked (cross-linked) together. Cross-linked polymers are hard, rigid and brittle because of their network structure.

Example: Bakelite, urea-formaldehyde resin etc.

MOLECULAR MASS OF POLYMERS

Molecular weight of any polymer is one of the important parameters as it links directly to the physical properties of polymers. In general, higher the molecular mass, tougher is the polymer.

During the formation of polymers, different polymers have different degrees of polymerization i.e., they have different chain lengths. Thus, molecular masses of individual macromolecules in a particular sample of polymer are different. Hence, an average value of the molecular mass is taken. There are two ways through which average molecular masses can be calculated.

Number – average molecular mass ($\bar{M}_n$)

Mass – average molecular mass ($\bar{M}_w$)

NUMBER – AVERAGE MOLECULAR MASS

Number – average molecular mass is the mass obtained when total mass of all the molecules of a sample is divided by the total number of molecules.

Example, in a particular sample,

N_1 molecules have molecular mass M_1 each.

N_2 molecules have molecular mass M_2 each.

N_3 molecules have molecular mass M_3 each and so on.

Then, we have

Total mass of all the N_1 molecules = N_1M_1

Total mass of all the N_2 molecules = N_2M_2

Total mass of all the N_3 molecules = N_3M_3 and so on.

Therefore, total mass of all the molecules = $N_1M_1 + N_2M_2 + N_3M_3 = \sum N_iM_i$

Total number of all the molecules = $N_1 + N_2 + N_3 = \sum N_i$

Hence the number – average molecular mass is given by

$$\bar{M}_n = \frac{N_1M_1 + N_2M_2 + N_3M_3 + \dots}{N_1 + N_2 + N_3 + \dots}$$

$$\bar{M}_n = \frac{\sum N_iM_i}{\sum N_i}$$

M_n is usually determined by osmotic pressure method.

MASS – AVERAGE MOLECULAR MASS

Mass – average molecular mass is the mass obtained when sum of the products of total mass of groups of molecules (Having different molecular masses) and their respective molecular masses is divided by total mass of all the molecules.

For example, in a particular sample,

N_1 molecules have molecular mass M_1 each.

N_2 molecules have molecular mass M_2 each.

N_3 molecules have molecular mass M_3 each and so on.

Then, we have

Total mass of all the N_1 molecules = N_1M_1

Total mass of all the N_2 molecules = N_2M_2

Total mass of all the N_3 molecules = N_3M_3 and so on.

The products of total mass of different groups of molecules with their respective molecular masses will be $(N_1M_1 \times M_1)$, $(N_2M_2 \times M_2)$, $(N_3M_3 \times M_3)$, etc.

That is, $N_1M_1^2$, $N_2M_2^2$, $N_3M_3^2$, etc.

Sum of the products = $N_1M_1^2 + N_2M_2^2 + N_3M_3^2 + \dots$

$$= \sum N_i M_i^2$$

Total mass of all the molecules = $N_1M_1 + N_2M_2 + N_3M_3 + \dots = \sum N_i M_i$

Hence, mass – average molecular mass is given by

$$\overline{M}_w = \frac{N_1M_1^2 + N_2M_2^2 + N_3M_3^2 + \dots}{N_1M_1 + N_2M_2 + N_3M_3 + \dots}$$

$$\overline{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i}$$

$\overline{M}_w$ is generally obtained by techniques like ultra centrifugation or sedimentation.

Poly dispersity index (PDI)

In general, we have two different types of polymeric substances

- 1) Polymers whose molecules have same or narrow range of molecular masses. Such polymers are said to be monodisperse.
- 2) Polymers which have wide range of molecular masses. Such polymers are called polydisperse.

In order to have an idea about the homogeneity of a polymer, we use the term called polydispersity index. It is the ratio of the mass average molecular mass to the number average molecular mass.

For natural polymers, PDI is usually equal to one showing that they are monodisperse. Such polymers are homogeneous. Synthetic polymers generally have $PDI > 1$ showing that they are less homogeneous and polydisperse.

NUMERICAL PROBLEMS

1. In a sample of a polymer, 100 molecules have molecular mass 10^3 g/mol, 250 molecules have molecular mass 10^4 g/mol, 300 molecules have molecular mass 10^5 g/mol, calculate the number average, weight average molecular mass of the polymer, PDI and comment on it.

It is given that,

$$N_1 = 100 \text{ \& } M_1 = 10^3 \text{ g/mol,}$$

$$N_2 = 250 \text{ \& } M_2 = 10^4 \text{ g/mol,}$$

$$N_3 = 300 \text{ \& } M_3 = 10^5 \text{ g/mol,}$$

The number average molecular mass of the polymer is given by

$$\begin{aligned}\bar{M}_n &= \frac{\sum N_i M_i}{\sum N_i} = \frac{N_1 M_1 + N_2 M_2 + N_3 M_3 + \dots}{N_1 + N_2 + N_3 + \dots} \\ \bar{M}_n &= \frac{100 \times 10^3 + 250 \times 10^4 + 300 \times 10^5}{100 + 250 + 300} \\ &= \frac{(1 + 25 + 300) \times 10^5}{650} = 0.501 \times 10^5 = 50100 \text{ g/mol}\end{aligned}$$

The weight average molecular mass of the polymer is given by

$$\overline{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i} = \frac{N_1 M_1^2 + N_2 M_2^2 + N_3 M_3^2 + \dots}{N_1 M_1 + N_2 M_2 + N_3 M_3 + \dots}$$

$$\overline{M}_w = \frac{100 \times (10^3)^2 + 250 \times (10^4)^2 + 300 \times (10^5)^2}{100 \times 10^3 + 250 \times 10^4 + 300 \times 10^5}$$

$$= \frac{(1 + 250 + 30000) \times (10^8)}{(1 + 25 + 300) \times 10^5} = \frac{30251 \times 10^3}{326}$$

$$= 92794 \text{ g/mol}$$

Poly dispersity index, $PDI = \frac{\overline{M}_w}{\overline{M}_n} = \frac{92794}{50100} = 1.85$

PDI is greater than one. Therefore, the given polymer is less homogeneous and polydisperse in nature.

2. In a polymer sample, 20 % of molecules have molecular mass 15000 g/mol, 35 % molecules have molecular mass 25000 g/mol, remaining molecules have molecular mass 20000 g/mol, calculate the number average, weight average molecular mass of the polymer, and PDI.

It is given that,

$$N_1 = 20 \text{ \& } M_1 = 15000 \text{ g/mol,}$$

$$N_2 = 35 \text{ \& } M_2 = 25000 \text{ g/mol,}$$

$$N_3 = 45 \text{ \& } M_3 = 20000 \text{ g/mol,}$$

The number average molecular mass of the polymer is given by

$$\overline{M}_n = \frac{\sum N_i M_i}{\sum N_i} = \frac{N_1 M_1 + N_2 M_2 + N_3 M_3 + \dots}{N_1 + N_2 + N_3 + \dots}$$

$$\overline{M}_n = \frac{20 \times 15000 + 35 \times 25000 + 45 \times 20000}{20 + 35 + 45}$$

$$= \frac{(300 + 875 + 900) \times 10^3}{100} = 20750 \text{ g/mol}$$

The weight average molecular mass of the polymer is given by

$$\overline{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i} = \frac{N_1 M_1^2 + N_2 M_2^2 + N_3 M_3^2 + \dots}{N_1 M_1 + N_2 M_2 + N_3 M_3 + \dots}$$

$$\overline{M}_w = \frac{20 \times (15000)^2 + 35 \times (25000)^2 + 45 \times (20000)^2}{20 \times 15000 + 35 \times 25000 + 45 \times 20000}$$

$$= 21385.54 \text{ g/mol}$$

$$\text{Poly dispersity index, PDI} = \frac{\overline{M}_w}{\overline{M}_n} = \frac{21385}{20750} = 1.03$$

Calculate weight average molecular weight of polyvinyl chloride (PVC), if degree of polymerization is 12250.

The structure of repeat unit in PVC is $-(\text{CH}_2-\text{CHCl})-$

Molecular mass of repeat unit is = 62.5 g/mol

Degree of polymerization is = 12250

We know that,

$$\text{Degree of polymerization} = \frac{\text{Molecular mass of polymer}}{\text{Molecular weight of repeating unit}}$$

$\therefore$ Molecular mass of polymer = Degree of polymerization $\times$ Molecular weight of repeating unit

$$= 62.5 \times 12250$$

$$= 7.66 \times 10^5 \text{ g/mol}$$

CONDUCTING POLYMERS

Polymers are best known for their effectiveness as electrical insulators. Indeed electrical wiring throughout the world is now sheathed in plastic. In 1977, a chemist Shirakawa accidentally added a catalyst 1000 times more than required amount during polymerization of polyacetylene, which resulted in conducting polyacetylene. There are now numerous polymers with substantial electrical conductivities and the topic of electrically conducting polymers still continues to excite scientific community across the globe. In future, it may lead to a revolution when low cost conducting polymers with conductivity similar to that of copper or aluminum are synthesized.

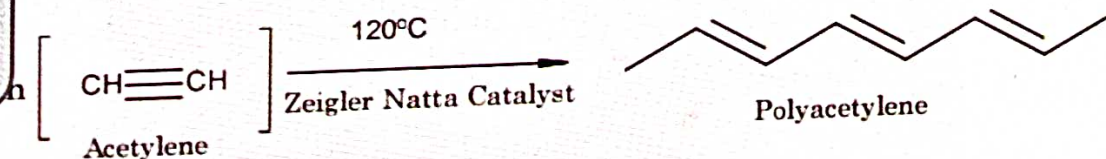
Conducting polymers are classified as extrinsic and intrinsic conducting polymers. Extrinsic conducting polymers are prepared by mixing polymers with conducting fillers like metal fibers, metal oxides, carbon black, etc. Here, conductivity is not due to the matrix polymer but due to the added conducting fillers. In intrinsic conducting polymers, conductivity is due to organic polymers themselves which start conducting upon doping with suitable agents.

Organic polymers having electrical conductivity in the order of a conductor are called as conducting polymers (Also as inherently or intrinsically conducting polymers or synthetic metals).

Few polymers which conduct electricity upon doping are polyacetylene, polyaniline, polypyrrole, polythiophene, polyphenylene, etc.

PREPARATION OF POLYACETYLENE

Polyacetylene films are synthesized using the Zeigler-Natta catalyst, The catalyst tetrabutoxytitanium and triethylaluminum suspended in silicone oil through which acetylene is passed, is stirred for two hours at 120° C, and then cooled slowly to room temperature resulting in thin sheets of polyacetylene.



An organic polymer to conduct electricity upon doping, should have

Linear structure with relatively rigid chains.

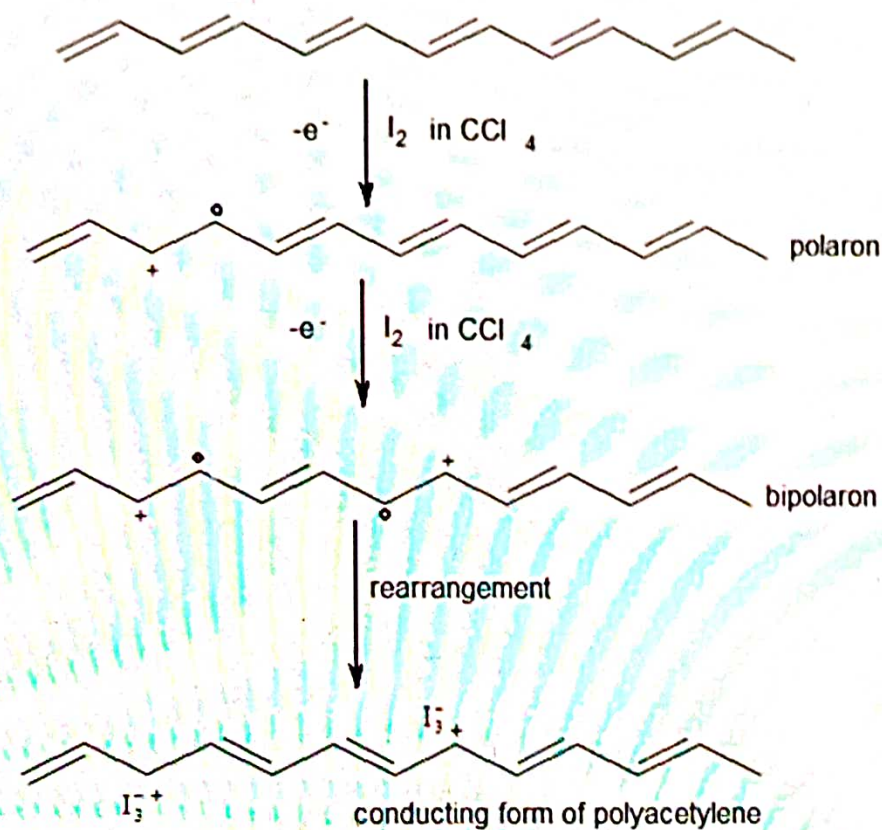
Extensively conjugated polymeric back bone (π back bone).

Above kind of polymers in their original form are insulators. But on suitable doping, their conductivity can be increased considerably. For instance, conductivity on polyacetylene is 4.4×10^{-5} S/cm, whereas, that of doped polyacetylene is 400 S/cm. Doping creates charge carriers like holes or excess electrons (similar to doped silicon) and thereby, an organic polymer becomes a conductor. The important doping techniques are:

1. Oxidative doping (p-type doping): Here the polymer is treated with an oxidizing agent like iodine, aluminium chloride, perchloric acid, etc. Oxidizing agent takes away electron from π -back bone of the chain creating a hole. Thus polymer becomes p-type conductor as it conducts by the movement of holes.

MECHANISM OF CONDUCTION IN POLYACETYLENE ON OXIDATIVE DOPING

The polymer chain is partially oxidized by using a mild oxidizing agent such as I_2 in CCl_4 . In the first step, one electron is removed from π -backbone of the chain leaving a positive charge on one of the carbon atoms. One more electron left out on another carbon makes it a free radical. The free radical-cation ion thus formed is called as polaron. Similarly, on second oxidation, another free radical-cation is formed which is called as bipolaron. The radicals migrate towards each other and combine to form one double bond. The chain now have two positive charges (holes). Positive charges are compensated by I_3^- ions and thus the polymer is electrically neutral. As positive charges are mobile, on applying potential they move and thus making a conductive polymer.



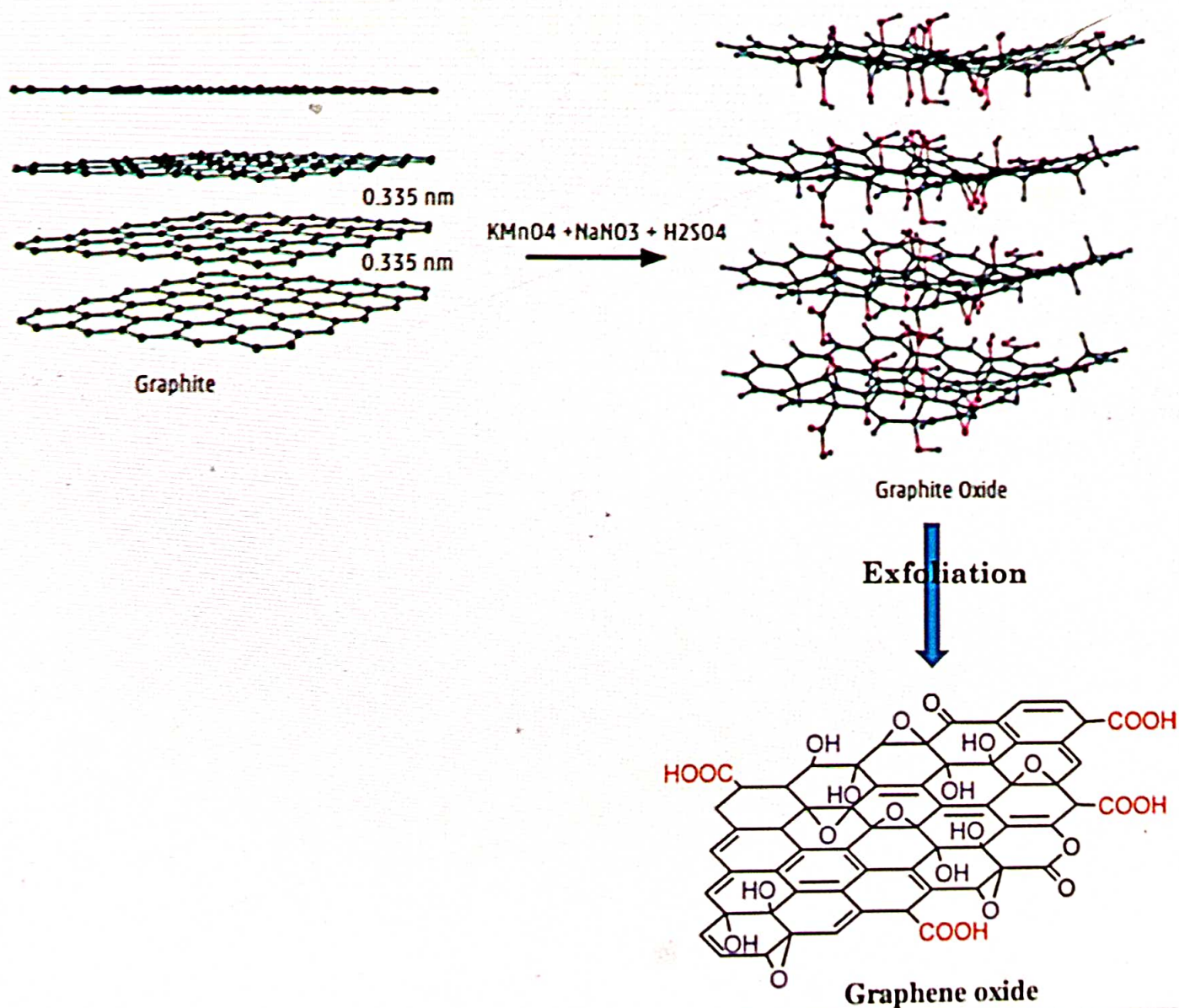
2. Reductive doping (n-type doping): Here the polymer is treated with a reducing agent like sodium or lithium naphthalides. Reducing agent supplies electrons to π -back bone of the chain. Thus polymer becomes n-type conductor as it conducts by free mobile electrons.

MECHANISM OF CONDUCTION IN POLYACETYLENE ON REDUCTIVE DOPING

In this case, polymer chain is partially reduced by using a mild reducing agent such as Na-Napthalide in tetrahydrofuran (THF) as solvent. In the first step, one electron is added to polymer chain forming a free radical-anion which is called as polaron. Similarly, in the second step another free radical-anion is formed which is called as bipolaron. The radicals migrate towards each other and combine to form one double bond. This bipolaron on rearrangement gives a conducting polymer with two negative charges as carriers of electric current. Negative charges are compensated by Na^+ ions and polymer as a whole is electrically neutral. Since negative charges are the carriers of electric current, this type of doping is called as n-doping.

Graphene Oxide (GO) is an oxidized derivative of graphite. It is obtained from cheap and abundantly available graphite by controlled oxidation. There are several methods available for the preparation of graphene Oxide. Different oxidizing agents are used in each method. **Hummers method** developed by Hummers and Offeman in 1958, is a relatively safe method for preparing graphite oxide. In this method graphite is subjected to reaction with an anhydrous mixture of concentrated sulfuric acid, sodium nitrate, and potassium permanganate. The oxidation reaction is completed within 2 h at a temperature below 45°C.

Interlayer distance between adjacent layers in graphite is 0.335 nm. Graphite is hydrophobic and its layers cannot be easily exfoliated. But, in GO interlayer distance between adjacent layers increases to 0.625 nm after introducing epoxides, alcohols, ketone carbonyls, and carboxylic groups. These oxygenated functionalities weaken the van der Waals interactions between the layers of graphite oxide and make GO more hydrophilic. Hydrolysis of GO in the presence of polar solvent dimethylformamide (DMF) and sonication results in complete exfoliation in water, yielding individual graphene oxide sheets.



Properties of Graphene Oxide

1. The structure and chemical composition of GO is not fixed, it can be fine-tuned by varying degree of oxidation and reduction. Completely oxidized GO is a yellow solid with C:O ratio between 2.1 and 2.9. It retains the layer structure of graphite but with a much larger and irregular spacing. Few of the important properties of GO are:
2. Graphene Oxide (GO) is more hydrophilic due to presence of polar functional groups. It can be dispersed in water and other polar solvents.
3. GO exhibits tunable electrical and optical properties. Reduction of GO increases its electrical conductivity. The band gap of GO can be varied from 0.5 to 2.2 eV depending on the synthesis procedure and post synthesis reduction procedure. GO can be transformed from insulator to semiconductor and further to a semimetal.
4. GO exhibits giant optical nonlinearity. Nonlinear optical (NLO) properties of GO depends on band gap. Thus, it is possible to tune the optical nonlinearity of GO by tailoring the band gap through reduction.
5. GO exhibits high specific surface area and highly effective in desalination of water and removal of toxic heavy metals from water.
6. GO is bio compatible and can be functionalized easily.

Applications of Graphene Oxide

Few important applications of GO related to its properties are:

1. Due to tunable electrical and optical properties, GO is a useful candidate for various applications like energy storage devices, transparent conducting electrodes, photodetectors, light-emitting devices, all-optical switching and fast optical communications.
2. Graphite oxide can be used for the large-scale production of graphene by chemical reduction.
3. GO membranes can be used for cost effective water purification.
4. Due to its bio compatibility and functionalization ability, GO is found to be useful as sensors, biosensors, effective components for medical applications, including transplant devices, invasive instruments, and implants.

SOLAR ENERGY

INTRODUCTION

Solar radiation is believed to be the only realistic alternative to fossil fuel. It is a clean and renewable source of energy available in abundance. As per an estimate, solar energy received by earth surface per day if converted into electrical energy is sufficient to meet energy requirements of whole world for 16 years.

SOLAR PHOTOVOLTAIC CELL (PV CELL)

Solar photovoltaic cell (PV cell) is a device used to convert solar light directly into an electric current. It is based on photovoltaic effect. According to which, when a light particle with energy $E=hu$ falls on a material and if the energy of photon is greater than binding energy of electron then it can eject an electron from it creating an electron hole pair.

In solar PV cell, when solar light falls on the material, electrons are ejected producing electron-hole pair. Then, electrons are separated from holes and made to move through external circuit to recombine with holes. This movement of electrons through an external circuit generates electric current.

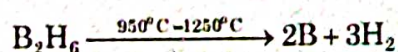
Entire spectrum of solar energy from IR to UV is in the range of 0.5 eV to 2.9 eV. There are several materials with their binding energy less than solar energy and in principle can be used in photovoltaic cells. More than 90% of today's solar cells are based on crystalline (mono- and polycrystalline) silicon. First solar cell based on crystalline silicon p-n junction was discovered by Bell Telephone laboratories in the early 1950s with a power efficiency of 6%. This discovery came after more than a century of the discovery of photovoltaic effect by Henri Becquerel (1839). After decades of research and development (R&D), efficiency of Si solar cell has presently reached 24% in the laboratory and 16% at the commercial scale. Besides the crystalline Si solar cells, thin film telluride (CdTe), $\text{CuIn}_{(1-x)}\text{Ga}_x\text{Se}_2$ based solar cells are currently used for power generation with laboratory-scale power efficiencies at 16% and 20%, respectively. Power

efficiency is defined as the ratio of the output electrical power to the inflowing solar power under standard conditions.

DOPING OF Si BY DIFFUSION TECHNIQUE

Si is semiconductor and its conductivity can be increased by doping with minute amount of either trivalent or pentavalent atoms. Doping is commonly carried out by diffusion technique. It is a chemical process in which dopant sources are decomposed thermally at temperatures between 950°C and 1250°C in a tube furnace. Dopant atoms thus formed thermally diffuses into silicon substrate. Typical dopant sources used are BBr₃, POCl₃ (liquids) B₂O₃, P₂O₅ (solids) PH₃, B₂H₆ gases.

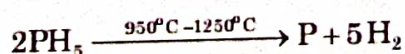
p-doping: In case of p-doping, a required quantity of dopant source like B₂H₆ is subjected to pyrolysis in a reactor containing Si substrate. Trivalent boron atoms formed by decomposition, diffuse in to the lattice of Si.



When trivalent atom is put into the lattice of tetravalent Si, it shares three of its valence electrons with Si atoms forming covalent bonds. Since trivalent atom does not have fourth electron to share with silicon, fourth covalent bond between Si and trivalent atom contains only one electron, thus generating a hole in place of other electron. Each dopant atom introduced into lattice generates a hole. These holes are responsible for electrical conductivity. Therefore, this type of semiconductor is called as p-type semiconductor.

A p-type semiconductor is obtained by doping Si with neutral trivalent dopant atoms, therefore as a whole it is neutral and does not carry positive charge.

n-doping: In case of n-doping, a required quantity of dopant source like PH₃ is subjected to pyrolysis in a reactor containing Si substrate. Pentavalent phosphorous atoms formed by decomposition, diffuse in to the lattice of Si.

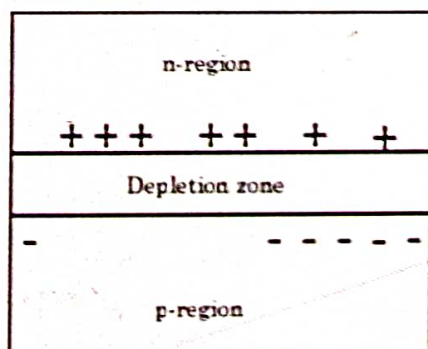


When pentavalent atom is put into the lattice of tetravalent Si, it shares four of its valence electrons with Si atoms forming covalent bonds. Fifth electron of pentavalent atom remains unbonded. Thus, each dopant atom introduced into lattice contributes an extra electron. These electrons are responsible for electrical conductivity. Therefore, this type of semiconductor is called as n-type semiconductor.

An n-type semiconductor is obtained by doping Si with neutral pentavalent dopant atoms, therefore as a whole it is neutral and does not carry negative charge.

p-n junction diode:

The device used in photovoltaic cell is a p-n junction diode which is obtained by the combination of p-type and n-type semiconductors. When a p-type semiconductor is brought in contact with an n-type of semiconductor, electrons from n-type diffuse towards p-type to occupy the holes. This results in electron-hole recombination at the vicinity of junction. Hence this region gets depleted and will not have any charge carriers. This region is called as depleted region.



p-n junction diode

During diffusion, n-region loses electrons and becomes positively charged & p-region gains electrons and becomes negatively charged. Hence an electrical double layer is formed at the junction of p-type and n-type semiconductors. The positive charges present on n-side oppose the movement of hole from p-type to n-type side. Similarly, the negative charges present on p-side oppose movement of electrons from n-type to p-type side. Due to this charge separation, there exist a potential across the junction, which is called as barrier potential. That is, to make the further movement of electrons, a potential more than the barrier potential should be applied. Thus, on applying a potential more than the barrier potential, electrons can move in only one direction from

p-type to n-type side through p-n junction. The flow of electrons in reverse direction is forbidden. This property of a p-n junction diode is used to separate electrons from holes.

Main requirements for a material to be used in a solar PV cell are:

1. It should eject electrons by absorbing sunlight creating electron-hole pair.
2. Ejected electrons should be separated from holes and made to move through external circuit to recombine with holes. This movement of electrons through an external circuit generates electric current.

A p-n junction diode satisfies both these conditions;

1. p- side of diode is exposed to solar light which ejects electrons creating electron-hole pair.
2. These ejected electrons move to n-side through p-n junction by overcoming the barrier potential. Since, p-n junction allows only one way movement of electrons from p- to n-side; these electrons are separated from holes. Now, these electrons must flow through the external circuit to recombine with holes present in p-side of semiconductor.

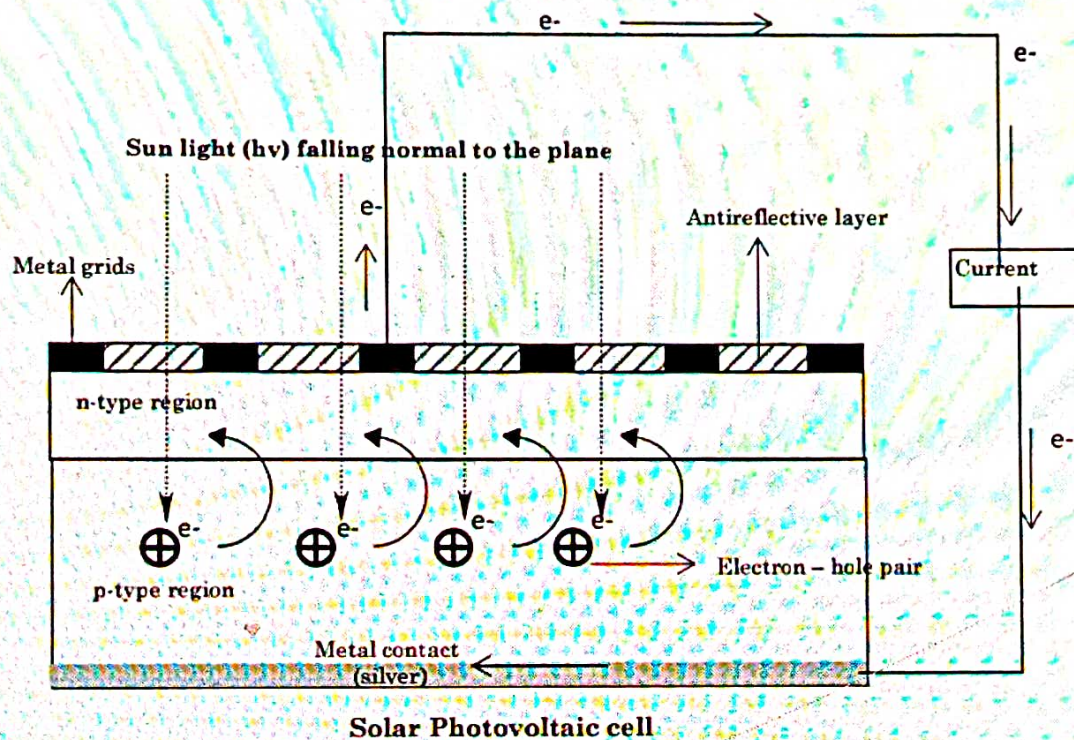
CONSTRUCTION AND WORKING OF A SOLAR PHOTOVOLTAIC (PV) CELL

Photovoltaic cell mainly consists of a p-n junction diode made from Si semiconductor. It has two electrical contacts; one is a metallic grid over n-type and second is a layer of silver metal at the back of p-type semiconductor. An antireflective layer (Silicon nitride or TiO_2) is coated in between the metal grids to prevent reflection of solar light.

When sunlight with their photons carrying energy $E=h\nu$ (which is enough to cross the barrier potential), falls on p-n diode, electrons are ejected from p-type semiconductor creating an electron-hole pair. These ejected electrons move to n-side through p-n junction. Since p-n junction allows only one way movement of electrons, these electrons must flow through the external circuit to recombine with holes present

in p-side of semiconductor. This movement of electrons from n-side to p-side through the external circuit generates an electric current.

Electrical energy obtained from a PV cell is directly proportional to area of the cell exposed to light radiation and intensity of light radiation which can eject electrons.



Advantages of solar cells

1. They are environmental friendly.
2. It is non-polluting, low maintenance energy source.
3. They need no recharging.
4. They do not suffer wear and tear as they have no movable parts.
5. They do not corrode.
6. They operate at ambient temperature.
7. Power generation by PV cell can minimize dependence on fossil fuel and at the same time reduce greenhouse gas emissions mainly CO_2 .

(Approximately, 8182 ton of greenhouse gases can be avoided entering into the atmosphere each year from a 5MW capacity PV power plant).

Disadvantages of solar PV cell

1. Power efficiency of a commercial Si based PV cell is very low(16%).
2. Silicon required for PV cell is of high purity (99.999%). High cost of Si purification has resulted in higher price of crystalline silicon solar cells.

A Little More...

Future of Solar energy

Lower power efficiency and higher cost of purifying silicon are considered the main hurdles in the development of PV cell. Because of these problems, contribution of PV power generation to the overall energy supply is limited. In spite of its exponential growth in the 2000s, solar energy contributes just about 0.1% to overall energy requirements. To increase this contribution, prices of PV cell must fall significantly. Using of thin film-based technologies is believed to reduce the production cost of PV modules. Solar PV cells based on cadmium telluride (CdTe), $\text{CuIn}_{(1-x)}\text{Ga}_x\text{Se}_2$ have favorable efficiency and lower cost than conventional crystalline Si based PV cells. But these technologies are hindered by limited resources for their manufacture.

Two more important types of solar cells, namely, organic solar cell and dye-sensitized solar cell (DSC) have been developed recently but not yet commercialized. In these cells conjugated molecular species such as conducting polymers, molecules and dyes, which are capable of absorbing light and conducting charge and thereby acting as organic semiconductors are used. These cells have combined advantage of the lightweight, flexibility and easy fabrication of organic molecules with the low module cost. Organic solar cell is a simple bilayer device in which an electron-donating layer and an electron-accepting layer of two different organic semiconductors form a planar junction at which light-generated excitons dissociate into charge carriers and are collected at the electrodes. This type of cell consisting a bilayer copper phthalocyanine/perylene-tetracarboxylic derivative (CuPc/PTC) was first invented by Tang (1986), with a power conversion efficiency of 0.95%. Currently, the best performing bilayer cells have efficiencies of around 5%, and contain a vacuum-deposited molecular layer of a

HYDROGEN AS GREEN FUEL

More than 90 % of the global energy requirements are being met by carbon containing fuels. They are used mainly in electricity production, transportation, industrial and domestic heating. These fuels are non-renewable and emit carbon dioxide which is the main greenhouse gas. Therefore, developing an alternative source of energy which is clean, continuous and renewable, is required to meet global energy

demand. Hydrogen is the only promising alternative fuel to carbonaceous fuels. The main advantages of hydrogen as fuel are:

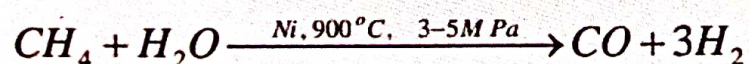
1. Hydrogen is an ideal, highly efficient, renewable, clean, and sustainable energy source.
2. It is abundant from various sustainable sources (biomass or water).
3. Energy content of hydrogen is 122 kJ/g, which is 2.75 times greater than hydrocarbon fuels like petrol and Diesel. However, its volumetric density is lower than petrol, diesel.
4. The combustion product is water, which is not a pollutant.
5. It can be used as fuel in Fuel cell for production of electric current.
6. It can be used as chemical fuel and burnt directly to produce heat energy.
7. It has high storage capability, thus considered as an ideal alternative source of energy for fossil fuels.

There are few drawbacks of hydrogen as a fuel, which need to be addressed before it is used as a fuel. Main drawbacks are:

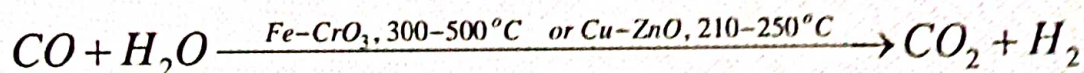
1. Hydrogen is a colourless, odourless gas, which is extremely flammable and highly explosive. Therefore, storing, handling, transportation is very difficult.
2. Cost of storing of hydrogen as a compressed gas or liquid is high as it requires lot of energy.
3. Cost of development of hydrogen infrastructure (hydrogen technology) is very high.
4. Hydrogen is a secondary source of energy. Production of clean hydrogen from renewable source is a major challenge.

Production of hydrogen

Currently, 96% of hydrogen is being produced from Steam Reforming of Natural Gas and the partial oxidation of heavy hydrocarbons or Coal.



About 12 % of CO produced in the process is further converted to hydrogen and carbon dioxide through water gas shift reaction.



Thus the current hydrogen generation techniques rely on fossil fuels, require harsh conditions, with higher cost of production. Therefore, alternative methods that utilize renewable sources of energy for hydrogen production are being explored. The following two strategies are employed for production of clean hydrogen.

1. Electrolysis of water
2. Solar water splitting.

1. ELECTROLYSIS OF WATER

Majority of the hydrogen is produced today by steam reforming of hydrocarbons and it is being used for industrial applications. For few applications which require pure carbon free hydrogen, it is produced by water electrolysis. Water is the natural source of molecular hydrogen and water electrolysis is a simple water-splitting process that can be operated at room temperature. The energy required to extract 1 mol of hydrogen from water is about four times larger than that required to extract 1 mol of hydrogen from steam reforming of hydrocarbons (methane). 237.22 kJ/mol of energy and potential of 1.3 V is required to split water in to hydrogen and oxygen by electrolysis. Therefore, as of today, electrolysis cannot be used to produce green hydrogen required for energy storage in the form of chemical fuel. But, hydrogen can be produced by water electrolysis when cheap electricity is available. If more than the required amount of electric current is produced from solar energy, then the excess current can be used for water electrolysis and store the energy in hydrogen in chemical form.

POLYMER ELECTROLYTE MEMBRANE WATER ELECTROLYSIS

There are three components in a polymer electrolyte membrane water electrolysis cell.

POLYMER ELECTROLYTE MEMBRANE WATER ELECTROLYSIS

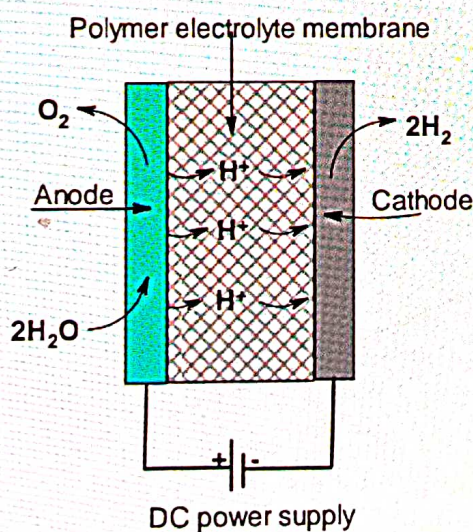
There are three components in a polymer electrolyte membrane water electrolysis cell.

Anode: Iridium metal particles dispersed on porous carbon is used as anode electrocatalyst for Oxygen evolution reaction.

Cathode: Platinum particles coated on porous carbon is used as cathode electrocatalyst for hydrogen evolution reaction.

Porous metal electrodes pressed on either sides of the surface of polymer membrane are used as anode and cathode.

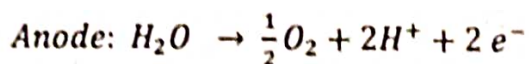
Electrolyte/Separator: A porous solid polymer electrolyte made of chemically stable sulfonated tetrafluoroethylene-based fluoropolymer (NAFION) is used as an electrolyte as well as a separator. It is a good ionic conductor for protons and bad electronic conductor. It prevents the spontaneous recombination of H_2 and O_2 .



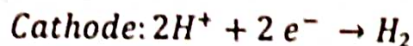
Polymer Electrolyte Membrane electrolysis

Working of cell

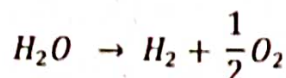
Water electrolysis is a non-spontaneous chemical reaction. It is driven by external electricity and by applying a sufficient voltage between the two end electrodes. Anode is connected to positive end of direct current and cathode is connected to negative end. Deionized water is circulated in the anodic chamber where it is oxidized liberating oxygen gas and hydrogen ions.



Hydrogen ions migrate through the Solid Polymer Electrolyte membrane to the cathode, where they are reduced into molecular hydrogen.



Overall reaction is:



Advantages of PEM electrolysis:

Use of polymer membrane avoids use of liquid acid electrolyte. Therefore, chance of electrolyte leakage is prevented.

Polymer membrane is chemically stable and non-corrosive.

Pure carbon free hydrogen is obtained from this method.

Excess cheap current from renewable sources like solar power can be converted to Hydrogen gas and stored as chemical energy.

ALKALINE WATER ELECTROLYSIS

There are four components in an alkaline water electrolysis cell.

Anode: Nickel metal particles dispersed on porous carbon is used as anode electrocatalyst for oxygen evolution reaction.

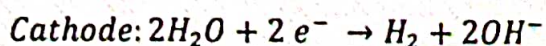
Cathode: Nickel metal particles coated on porous carbon is used as cathode electrocatalyst for hydrogen evolution reaction.

Electrolyte: Aqueous solution of potassium hydroxide (KOH) with 20 to 30 wt.% KOH is used as an electrolyte.

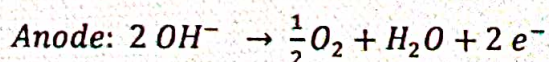
Separator: Porous dense anion exchange membrane is used as the separator. It is a good ionic conductor for hydroxide ions and bad electronic conductor. It prevents the spontaneous recombination of H_2 and O_2 .

Working of cell

Water electrolysis is a non-spontaneous chemical reaction. It is driven by external electricity and by applying a sufficient voltage between the two end electrodes. Anode is connected to positive end of direct current and cathode is connected to negative end. At the cathode, water molecules are reduced by electrons to hydrogen and negatively charged hydroxide ions.



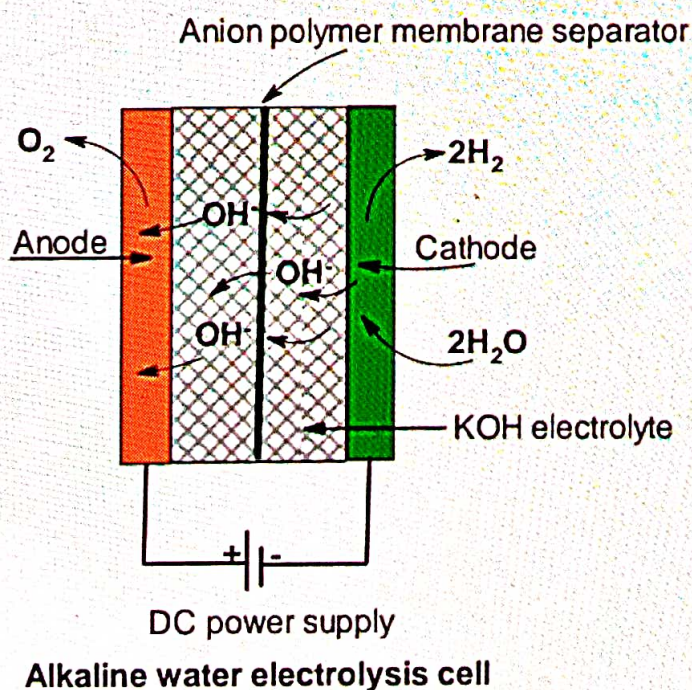
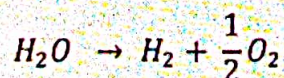
Hydroxide ions migrate through the alkaline KOH Electrolyte pass through separator membrane to the anode, where they are oxidized into oxygen.



At the anode, hydroxide ions are oxidized to oxygen and water while releasing electrons.

Overall, a water molecule splits in to hydrogen and oxygen in the ratio of 2:1.

Overall reaction is:



Advantages of PEM electrolysis:

Alkaline water electrolysis is an easier, cheaper, and simple method for hydrogen production.

Electrodes are made of cheaper nickel metal.

Pure carbon free hydrogen is obtained from this method.

Excess cheap current from renewable sources like solar power can be converted to Hydrogen gas and stored as chemical energy.